SKILL – 1

GIT VERSION CONTROL

Introduction :

Git Version Control is used to manage and track changes in project files. It helps developers work together by saving different versions of code safely. Using Git, we can create branches, make updates, and merge changes without losing data. This skill helps us understand how real so ware teams collaborate and manage their projects efficiently

TASKS & OUTPUTS:

```
MINGW64:/c/Users/hp/git-skill1

hp@LAPTOP-16BLNNG6 MINGW64 ~
$ mkdir git-skill1

hp@LAPTOP-16BLNNG6 MINGW64 ~
$ cd git-skill1

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1
$ git init
Initialized empty Git repository in C:/Users/hp/git-skill1/.git/

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git config --global user.name "Eswar"

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git config --gloabal user.email "g.eswarsairam7@gmail.com"
error: unknown option  gloabal'
usage: git config list [<file-option>] [<display-option>] [--includes]
   or: git config get [<file-option>] [<display-option>] [--includes] [--all] [-
-regexp] [--value=<pattern>] [--fixed-value] [--default=<default>] [--url=<url>]
 <name>
   or: git config set [<file-option>] [--type=<type>] [--all] [--value=<pattern>
] [--fixed-value] <name> <value>
   or: git config unset [<file-option>] [--all] [--value=<pattern>] [--fixed-val
ue] <name>
   or: git config rename-section [<file-option>] <old-name> <new-name>
   or: git config remove-section [<file-option>] <name>
   or: git config edit [<file-option>]
   or: git config [<file-option>] --get-colorbool <name> [<stdout-is-tty>]

Config file location
    --[no-]global        use global config file
    --[no-]system        use system config file
    --[no-]local         use repository config file
    --[no-]worktree      use per-worktree config file
    -f, --[no-]file <file>
                         use given config file
    --[no-]blob <blob-id> read config from given blob object

Action
    --get                get value: name [<value-pattern>]
    --get-all            get all values: key [<value-pattern>]
    --get-regexp         get values for regexp: name-regex [<value-pattern>]
    --get-urlmatch       get value specific for the URL: section[.var] URL
    --replace-all        replace all matching variables: name value [<value-pat
tern>]
    --add                add a new variable: name value
    --unset              remove a variable: name [<value-pattern>]
    --unset-all          remove all matches: name [<value-pattern>]
    --rename-section     rename section: old-name new-name
    --remove-section     remove a section: name
    -l, --list           list all
    -e, --edit           open an editor
    --get-color          find the color configured: slot [<default>]
    --get-colorbool      find the color setting: slot [<stdout-is-tty>]

Display options
```

```
MINGW64:/c/Users/hp/git-skill1
    --unset-all           remove all matches: name [<value-pattern>]
    --rename-section      rename section: old-name new-name
    --remove-section      remove a section: name
    -l, --list            list all
    -e, --edit            open an editor
    --get-color           find the color configured: slot [<default>]
    --get-colorbool       find the color setting: slot [<stdout-is-tty>]

Display options
    -z, --[no-]null       terminate values with NUL byte
    --[no-]name-only      show variable names only
    --[no-]show-origin    show origin of config (file, standard input, blob, com
mand line)
    --[no-]show-scope     show scope of config (worktree, local, global, system,
 command)
    --[no-]show-names     show config keys in addition to their values

Type
    -t, --[no-]type <type>
                          value is given this type
    --bool                value is "true" or "false"
    --int                 value is decimal number
    --bool-or-int         value is --bool or --int
    --bool-or-str         value is --bool or string
    --path                value is a path (file or directory name)
    --expiry-date         value is an expiry date

Other
    --[no-]default <value>
                          with --get, use default value when missing entry
    --[no-]comment <value>
                          human-readable comment string (# will be prepended as
needed)
    --[no-]fixed-value    use string equality when comparing values to value pat
tern
    --[no-]includes       respect include directives on lookup


hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git config --global user.email "g.eswarsairam7@gmail.com"

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=g.eswarsairam7@gmail.com
user.name=Eswar
```

```
MINGW64:/c/Users/hp/git-skill1
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=g.eswarsairam7@gmail.com
user.name=Eswar
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ touch main.java notes,text

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ echo "Hello Git" > main.java

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ echo "Git Skill 1 notes" > notes,txt

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        main.java
        notes,text
        notes,txt

nothing added to commit but untracked files present (use "git add" to track)

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git add .
warning: in the working copy of 'main.java', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'notes,txt', LF will be replaced by CRLF the nex
t time Git touches it

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git status
On branch master

No commits yet
```

```
MINGW64:/c/Users/hp/git-skill1
  (use "git add <file>..." to include in what will be committed)
        main.java
        notes,text
        notes,txt

nothing added to commit but untracked files present (use "git add" to track)

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git add .
warning: in the working copy of 'main.java', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'notes,txt', LF will be replaced by CRLF the nex
t time Git touches it

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   main.java
        new file:   notes,text
        new file:   notes,txt


hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git commit -m "Initial commit with main.java and notes.txt"
[master (root-commit) 056f46f] Initial commit with main.java and notes.txt
 3 files changed, 2 insertions(+)
 create mode 100644 main.java
 create mode 100644 notes,text
 create mode 100644 notes,txt

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ cat notes,text

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git mv notes,text notes.txt
git commit -m "Renamed notes,text to notes.txt"
[master 6b3c9ef] Renamed notes,text to notes.txt
 1 file changed, 0 insertions(+), 0 deletions(-)
 rename notes,text => notes.txt (100%)

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git status
On branch master
nothing to commit, working tree clean

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git log --stat
commit 6b3c9ef57d97823e8a51b3a33a9f5bc489d99c17 (HEAD -> master)
Author: Eswar <g.eswarsairam7@gmail.com>
Date:   Tue Mar 17 07:54:00 2026 +0530

    Renamed notes,text to notes.txt
```

```
MINGW64:/c/Users/hp/git-skill1
hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ cat notes,text

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git mv notes,text notes.txt
git commit -m "Renamed notes,text to notes.txt"
[master 6b3c9ef] Renamed notes,text to notes.txt
 1 file changed, 0 insertions(+), 0 deletions(-)
 rename notes,text => notes.txt (100%)

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git status
On branch master
nothing to commit, working tree clean

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git log --stat
commit 6b3c9ef57d97823e8a51b3a33a9f5bc489d99c17 (HEAD -> master)
Author: Eswar <g.eswarsairam7@gmail.com>
Date:   Tue Mar 17 07:54:00 2026 +0530

    Renamed notes,text to notes.txt

 notes,text => notes.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)

commit 056f46f5952549af971a16a4da7bbecc689d1b4a
Author: Eswar <g.eswarsairam7@gmail.com>
Date:   Tue Mar 17 07:52:30 2026 +0530

    Initial commit with main.java and notes.txt

 main.java  | 1 +
 notes,text | 0
 notes,txt  | 1 +
 3 files changed, 2 insertions(+)

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git ls-files
main.java
notes,txt
notes.txt

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git rm notes,txt
git commit -m "Removed duplicate file notes,txt"
rm 'notes,txt'
[master 18d71c3] Removed duplicate file notes,txt
 1 file changed, 1 deletion(-)
 delete mode 100644 notes,txt

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$ git ls-files
main.java
notes.txt

hp@LAPTOP-16BLNNG6 MINGW64 ~/git-skill1 (master)
$
```

Git Url: https://github.com/Eswar2400032797/git-skill1.git